Finite Difference Numerical Methods for Partial Differential Equations

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How this chapter fits together

1. A PDE describes a continuous field; a computer stores finitely many numbers.
2. Taylor series tell us how to replace derivatives by differences.
3. Applying those formulas to a PDE gives a difference equation on a grid.
4. Initial/boundary data complete the discrete problem.
5. Truncation error measures approximation; stability measures error growth.
6. We apply the framework to heat, wave, Laplace, and finite-element problems.

Learning goals

By the end, you should be able to:

- derive first- and second-derivative formulas and state their accuracy;
- build a finite-difference scheme from a PDE and its data;
- explain why explicit time steps have stability limits;
- use the heat, wave, and Laplace stencils correctly;
- distinguish consistency, stability, and convergence;
- describe the basic weak-form idea behind finite elements.

6.1 Introduction: three prototype PDEs

Class	Prototype	Typical behavior
Parabolic	$u_t = ku_{xx}$	Diffusion smooths temperature; solutions often decay toward equilibrium.
Hyperbolic	$u_{tt} = c^2 u_{xx}$	Waves propagate at finite speed; oscillations persist.
Elliptic	$u_{xx} + u_{yy} = 0$	Steady-state field; values are constrained by boundary data.

These are qualitative descriptions, not universal claims about every solution. The chapter develops numerical methods for these three examples.

Why compute numerically?

- Exact solutions may require infinite series or integrals.
- Complicated geometries and data can make analytic solutions impractical.
- A numerical method produces approximate values at selected points.
- The method must be assessed: a plausible-looking plot is not enough.

Central question: How do we turn a continuous PDE into reliable algebraic calculations?

The basic numerical workflow

1. Define the physical domain and the PDE.
2. Choose a spatial mesh and, when needed, a time step.
3. Approximate derivatives at mesh points.
4. Impose initial and boundary conditions.
5. Compute the unknown nodal values.
6. Check resolution, truncation error, and stability.

6.2 Mesh notation: what do the symbols mean?

For a one-dimensional interval $0 \le x \le L$, divide the interval into N equal pieces:

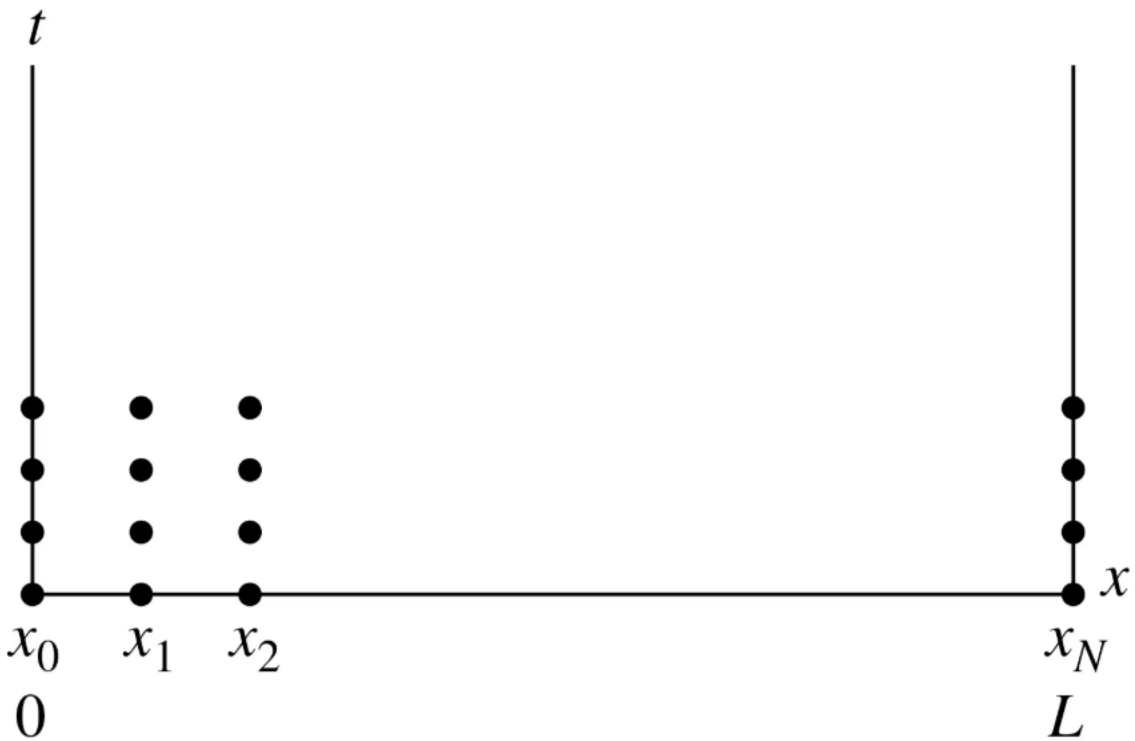
$$\Delta x = \frac{L}{N}, \qquad x_j = j\Delta x, \quad j = 0, 1, \ldots, N.$$

For a time-dependent problem, set

$$\Delta t > 0, \qquad t_m = m\Delta t, \quad m = 0, 1, 2, \ldots.$$

- j indexes space; m indexes time.
- $u(x_j, t_m)$ is the exact solution at a grid point.
- U_j^m denotes its numerical approximation.

Textbook figure: the space-time mesh



The explicit heat scheme will use three neighboring points at time t_m to compute one value at t_{m+1}. Cropped from Haberman, Chapter 6, Fig. 6.3.1 (printed p. 230).

Taylor's theorem: the source of the method

If f has $n+1$ continuous derivatives near x, then

$$f(x+h) = \sum_{r=0}^{n} \frac{h^r}{r!} f^{(r)}(x) + R_n,$$

where the remainder has the form

$$R_n = \frac{h^{n+1}}{(n+1)!} f^{(n+1)}(\xi)$$

for some intermediate point ξ between x and $x+h$.

- A small step h makes the omitted terms small when derivatives are bounded.
- The first omitted nonzero term predicts the leading error.

What does $\mathcal{O}(h^p)$ mean?

We write $E(h) = \mathcal{O}(h^p)$ if there are constants $C, h_0 > 0$ such that

$$|E(h)| \leq C|h|^p \qquad (0 < |h| < h_0).$$

- $\mathcal{O}(h)$: first-order error.
- $\mathcal{O}(h^2)$: second-order error.
- If the leading error behaves like Ch^p, halving h reduces it by about 2^p.

The constant C depends on the function and its derivatives; order notation describes scaling, not the exact error.

Forward difference: derive the formula

Taylor-expand at x:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(\xi).$$

Rearrange and divide by h:

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2} f''(\xi).$$

Therefore,

$$\boxed{f'(x) \approx \frac{f(x+h) - f(x)}{h}, \quad \text{error } \mathcal{O}(h).}$$

It uses the current point and the point one step forward.

Backward difference

Replace h by $-h$ in Taylor's theorem:

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(\xi).$$

Then

$$\boxed{f'(x) \approx \frac{f(x) - f(x-h)}{h}, \quad \text{error } \mathcal{O}(h).}$$

Forward and backward differences have the same formal order. Which one is convenient depends on which grid values are available, especially near boundaries.

Centered first difference: why it is more accurate

Expand in both directions:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \mathcal{O}(h^4),$$
$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \mathcal{O}(h^4).$$

Subtract; even powers cancel:

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3}f'''(x) + \mathcal{O}(h^5).$$

$$\boxed{f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \quad \text{error } \mathcal{O}(h^2).}$$

Compare first-derivative formulas

Formula	Approximation	Formal error
Forward	$[f(x+h)-f(x)]/h$	$\mathcal{O}(h)$
Backward	$[f(x)-f(x-h)]/h$	$\mathcal{O}(h)$
Centered	$[f(x+h)-f(x-h)]/(2h)$	$\mathcal{O}(h^2)$

The centered formula is often more accurate for smooth functions, but it is not always the best choice for every numerical scheme. In particular, stability and data availability matter.

Textbook example: approximate $f'(1)$

Question. Let $f(x) = \log x$ and $h = 0.1$. Approximate $f'(1)$ using the forward, backward, and centered differences. Compare with the exact derivative and compute the percentage error of each approximation.

Second derivative: derive the three-point formula

Add the Taylor expansions:

$$f(x+h) + f(x-h) = 2f(x) + h^2 f''(x) + \frac{h^4}{12} f^{(4)}(x) + \mathcal{O}(h^6).$$

Rearrange:

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{h^2}{12} f^{(4)}(x) + \mathcal{O}(h^4).$$

Hence

$$\boxed{f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}, \quad \text{error } \mathcal{O}(h^2).}$$

Weights and a useful check

The centered second derivative uses weights

$$\frac{1}{h^2}(1, -2, 1) \quad \text{at } x - h,\ x,\ x + h.$$

- The weights sum to zero: $1 - 2 + 1 = 0$.
- A derivative of a constant is zero, so a consistent derivative stencil must annihilate constants.
- The formula is exact for polynomials of degree at most three only insofar as their fourth derivative vanishes; in particular it is exact for quadratics and cubics.

Partial derivatives: hold the other variable fixed

For $u = u(x, y)$, an x-derivative changes x while y is fixed:

$$u_x(x_i, y_j) \approx \frac{U_{i+1,j} - U_{i-1,j}}{2\Delta x}.$$

Similarly, keeping x fixed,

$$u_y(x_i, y_j) \approx \frac{U_{i,j+1} - U_{i,j-1}}{2\Delta y}.$$

The same Taylor argument applies in each coordinate direction. For unequal spacings, keep Δx and Δy distinct.

The five-point approximation to the Laplacian

Apply the centered second-derivative formula in both coordinate directions:

$$u_{xx} \approx \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{(\Delta x)^2},$$

$$u_{yy} \approx \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{(\Delta y)^2}.$$

For equal spacing $h = \Delta x = \Delta y$,

$$\boxed{\nabla^2 u \approx \frac{U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1} - 4U_{i,j}}{h^2}.}$$

The spatial truncation error is $\mathcal{O}(h^2)$ for a sufficiently smooth solution.

Truncation error versus solution error

Local truncation error Insert the exact smooth solution into the discrete equation and measure the residual.

Global error Difference between the computed solution and the exact solution at grid points.

A small local truncation error is desirable, but does not alone guarantee a good computed solution. Errors can accumulate or grow. This motivates stability analysis.

Section 6.2 vocabulary: approximations and errors

Polynomial approximation Replace a function near a point by a low-degree polynomial matching its value and selected derivatives.

Taylor series with remainder An expansion of $f(x+h)$ into derivative terms plus a remainder that quantifies what was omitted.

Truncation error The remainder introduced when an infinite Taylor expansion is cut off after finitely many terms.

Finite-difference approximation A derivative approximation formed from function values at nearby grid points, without taking a limit.

Forward / backward difference First-derivative formulas using the point ahead of / behind the target point; each has error $O(h)$.

Centered difference A symmetric formula using points on both sides; the first-derivative formula has error $O(h^2)$.

Consistency In this section, the approximation is consistent if its truncation error tends to zero as the mesh spacing tends to zero.

Section 6.2 vocabulary: partial derivatives and the Laplacian

Partial derivative Differentiate with respect to one variable while holding the other variables fixed.

Centered second difference $f''(x) \approx [f(x+h) - 2f(x) + f(x-h)]/h^2$, with error $O(h^2)$ for a sufficiently smooth function.

Laplacian In two dimensions, $\nabla^2 u = u_{xx} + u_{yy}$; it measures the sum of the coordinate-wise curvatures.

Five-point stencil The grid pattern consisting of a node and its four horizontal/vertical neighbors used to approximate $\nabla^2 u$.

Weights of a difference formula The coefficients multiplying sampled values. For a consistent approximation to a derivative of a constant, these weights sum to zero.

These are the building blocks used in the heat, wave, and Laplace schemes that follow.

Section 6.2: key takeaways

- Taylor expansions derive the standard finite-difference formulas.
- Forward/backward first differences are first-order; centered first and second differences are second-order.
- Multidimensional formulas follow by applying one-dimensional formulas in each coordinate.
- The stencil weights for a derivative sum to zero.
- Accuracy is only one part of a reliable numerical method; stability comes next.

6.3 The one-dimensional heat equation

The model problem in Haberman is

$$\begin{cases} u_t = k u_{xx}, & 0 < x < L, \\ u(0,t) = u(L,t) = 0, & t > 0, \\ u(x,0) = f(x), & 0 \le x \le L, \end{cases} \qquad k > 0.$$

- $u(x,t)$ is temperature; k is thermal diffusivity.
- $f(x)$ gives the initial temperature profile.
- The boundary temperatures are held at zero.

Before discretizing: what is known and unknown?

- Known: the PDE, diffusivity k, initial profile f, and boundary values.
- Unknown: $u(x, t)$ for later times and interior positions.
- A numerical grid replaces the unknown function by nodal values U_j^m.
- The PDE gives a relation between neighboring nodal values at nearby times.

At the boundary, $U_0^m = U_N^m = 0$. At the initial time, $U_j^0 = f(x_j)$.

Derive FTCS: approximate the time derivative

At (x_j, t_m), use a forward difference in time:

$$u_t(x_j, t_m) \approx \frac{U_j^{m+1} - U_j^m}{\Delta t}.$$

Taylor's theorem gives

$$\frac{u(x_j, t_m + \Delta t) - u(x_j, t_m)}{\Delta t} = u_t(x_j, t_m) + \frac{\Delta t}{2} u_{tt}(x_j, \eta),$$

so the time approximation has error $\mathcal{O}(\Delta t)$. The unknown U_j^{m+1} is at the next time level.

Derive FTCS: approximate spatial curvature

Use the centered second difference at the current time:

$$u_{xx}(x_j, t_m) \approx \frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{(\Delta x)^2}.$$

Its truncation error is $\mathcal{O}((\Delta x)^2)$. Substitute the time and space approximations into $u_t = k u_{xx}$:

$$\frac{U_j^{m+1} - U_j^m}{\Delta t} = k \frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{(\Delta x)^2}.$$

Solve for the next time level

Define the dimensionless mesh ratio

$$s = \frac{k\Delta t}{(\Delta x)^2}.$$

Multiply the difference equation by Δt and isolate U_j^{m+1}:

$$\boxed{U_j^{m+1} = sU_{j-1}^m + (1-2s)U_j^m + sU_{j+1}^m.}$$

This is the forward-time, centered-space method, abbreviated FTCS.

FTCS: what the formula tells us

- One new interior value is computed from three old values.
- No linear system is solved at each time step: the method is explicit.
- The initial row is $U_j^0 = f(x_j)$; boundary values are reimposed at each time level.
- The scheme is first-order in time and second-order in space.

$$U_j^{m+1} = sU_{j-1}^m + (1-2s)U_j^m + sU_{j+1}^m.$$

A convex-average interpretation

If $0 \le s \le \frac{1}{2}$, then all three weights are nonnegative and

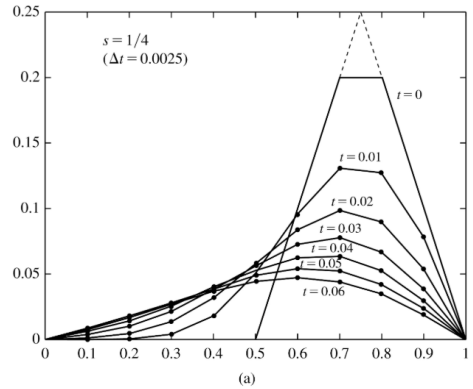
$$s + (1 - 2s) + s = 1.$$

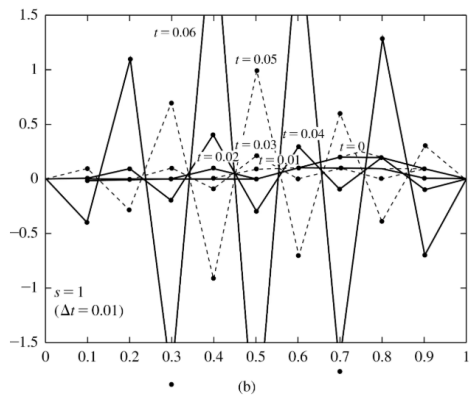
Thus the new value is a weighted average of the current value and its two neighbors. This helps explain why the stable range avoids some nonphysical oscillations. The convex-average observation is useful intuition; the Fourier analysis provides the standard stability condition.

In-class example: one heat-equation step

Question. Let $L = 1$, $N = 4$, $k = 1$, and choose Δt so that $s = 1/4$. The initial profile is $f(x) = \sin(\pi x)$ with zero endpoint values. Compute the first FTCS update at all interior nodes $j = 1, 2, 3$.

Textbook figure: stable and unstable heat computations





The textbook contrasts a computation with a stable mesh ratio and one with a mesh ratio that produces oscillatory growth. Stability is a property of the scheme together with the time/space step sizes. Cropped from Haberman, Chapter 6, Fig. 6.3.5 (printed p. 234).

FTCS local truncation error

Insert the exact solution into the FTCS difference equation. Taylor expansion gives a residual with leading terms proportional to $\Delta t u_{tt}$ and $k(\Delta x)^2 u_{xxxx}$:

$$\tau = \frac{\Delta t}{2} u_{tt} - \frac{k(\Delta x)^2}{12} u_{xxxx} + \mathcal{O}((\Delta t)^2 + (\Delta x)^4).$$

Therefore the local truncation error is generally

$$\boxed{\tau = \mathcal{O}(\Delta t) + \mathcal{O}((\Delta x)^2).}$$

The formula assumes sufficient smoothness to take the displayed derivatives. Different residual sign conventions may reverse the signs, but not the orders.

Consistency, stability, convergence

Consistency	The difference equation approaches the PDE as $\Delta t, \Delta x \to 0$; its local truncation error vanishes.
Stability	Small perturbations/errors do not grow without control under the discrete evolution.
Convergence	As the mesh is refined, the numerical solution approaches the exact solution.

For a well-posed linear initial-value problem, the Lax equivalence theorem connects consistency and stability to convergence. It is a theorem under hypotheses, not a blanket guarantee for every algorithm.

Fourier–von Neumann stability: why use a mode?

A linear constant-coefficient difference scheme can be tested by examining how it evolves oscillatory errors. Consider one Fourier mode

$$e_j^m = G^m e^{ij\theta},$$

where G is the amplification factor and θ is the spatial wave number in grid units.

- $|G| < 1$: this mode decays each time step.
- $|G| = 1$: its magnitude does not grow.
- $|G| > 1$: this mode grows and indicates instability.

Stability requires $|G| \leq 1$ for every admissible mode.

Derive the heat amplification factor

Insert $e_j^m = G^m e^{ij\theta}$ into the error recurrence:

$$Ge^{ij\theta} = se^{i(j-1)\theta} + (1-2s)e^{ij\theta} + se^{i(j+1)\theta}.$$

Divide by $e^{ij\theta}$ and use $e^{i\theta} + e^{-i\theta} = 2\cos\theta$:

$$G = 1 - 2s + 2s\cos\theta = 1 - 4s\sin^2(\theta/2).$$

Since $0 \le \sin^2(\theta/2) \le 1$, the smallest possible G is $1-4s$ for $s \ge 0$.

The FTCS stability condition

Require $-1 \le G \le 1$ for every Fourier mode. The upper bound is automatic for $s \ge 0$; the lower bound gives

$$1 - 4s \ge -1 \quad \Longrightarrow \quad s \le \frac{1}{2}.$$

Thus the standard 1-D FTCS stability restriction is

$$\boxed{0 \le \frac{k\Delta t}{(\Delta x)^2} \le \frac{1}{2}, \qquad \Delta t \le \frac{(\Delta x)^2}{2k}.}$$

Halving Δx requires reducing the maximum stable Δt by a factor of four.

What happens if the heat time step is too large?

- High-frequency spatial modes have $\sin^2(\theta/2)$ near one.
- If $s > 1/2$, their amplification factor can be less than -1.
- A negative amplification factor flips the sign each time step; magnitude above one makes the oscillation grow.
- The computed solution can become wildly oscillatory even when the physical heat solution should smooth out.

A smaller time step fixes this stability problem for FTCS; refining space alone while keeping Δt fixed can make it worse.

Heat equation as a random walk (interpretation)

When $0 \leq s \leq 1/2$, the update

$$U_j^{m+1} = sU_{j-1}^m + (1-2s)U_j^m + sU_{j+1}^m$$

can be read as a probability-weighted average: move left with probability s, move right with probability s, and stay with probability $1-2s$.

- The weights are nonnegative and sum to one.
- The update combines contributions from neighboring locations.
- This is an interpretation of the recurrence, not a claim that every temperature is itself a probability.

Separate variables in the difference equation

For insight into long-time behavior, try a product solution

$$U_j^m = \phi_j H_m.$$

Substitution into FTCS separates the time and space factors. The time factor has the form

$$H_{m+1} = \mu H_m \quad \Longrightarrow \quad H_m = \mu^m H_0.$$

The spatial factor satisfies a second-order difference eigenvalue problem with boundary values $\phi_0 = \phi_N = 0$. This parallels separation of variables for the continuous heat equation.

Discrete time modes: growth, decay, oscillation

For $H_{m+1} = \mu H_m$, the solution is $H_m = \mu^m H_0$.

- $0 < \mu < 1$: nonoscillatory decay.
- $-1 < \mu < 0$: alternating-sign decay.
- $\mu = -1$: pure sign alternation.
- $|\mu| > 1$: growth in magnitude; unstable behavior.

This simple recurrence explains the amplification-factor criterion $|G| \leq 1$ used in Fourier stability analysis.

Matrix form of the heat scheme

Collect the $N-1$ interior values into a vector

$$\mathbf{U}^m = (U_1^m, \ldots, U_{N-1}^m)^T.$$

Then

$$\mathbf{U}^{m+1} = A\mathbf{U}^m,$$

where

$$A = \begin{pmatrix} 1-2s & s & 0 & \cdots & 0 \\ s & 1-2s & s & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & s & 1-2s & s \\ 0 & \cdots & 0 & s & 1-2s \end{pmatrix}.$$

The zero boundary values are already incorporated into the interior system.

Why eigenvalues matter for stability

Repeated application gives

$$\mathbf{U}^1 = A\mathbf{U}^0, \quad \mathbf{U}^2 = A^2\mathbf{U}^0, \quad \mathbf{U}^m = A^m\mathbf{U}^0.$$

If $A\mathbf{v} = \mu\mathbf{v}$, then

$$A^m\mathbf{v} = \mu^m\mathbf{v}.$$

Thus eigenvalues describe the time behavior of special spatial modes. For a symmetric matrix, eigenvalues are real and eigenvectors can be chosen orthogonal. Stability requires that the relevant amplification eigenvalues do not have magnitude greater than one.

Nonhomogeneous heat equation

A source term changes the model to

$$u_t = k u_{xx} + F(x, t).$$

Using the same differences gives

$$U_j^{m+1} = s U_{j-1}^m + (1 - 2s) U_j^m + s U_{j+1}^m + \Delta t F(x_j, t_m).$$

The source contributes heat (or removes it if negative). The source term changes the update but does not remove the need to check stability of the underlying scheme.

Alternative heat schemes: why consider them?

FTCS is easy to implement but requires $\Delta t \leq (\Delta x)^2/(2k)$. If spatial accuracy requires a fine mesh, this can force many time steps. Implicit or partially implicit methods may allow larger time steps at the cost of solving equations at each step.

- Richardson's centered-time scheme: uses centered time differences but is unstable for the standard heat problem.
- Backward-time centered-space (BTCS): implicit; stable for all positive mesh ratios in the standard linear problem.
- Crank–Nicolson: averages spatial diffusion at the old and new times; second-order in time and unconditionally stable for the standard linear heat equation.

Backward-time centered-space (BTCS)

Approximate u_t by a backward time difference and u_{xx} at the new time:

$$\frac{U_j^{m+1} - U_j^m}{\Delta t} = k \frac{U_{j+1}^{m+1} - 2U_j^{m+1} + U_{j-1}^{m+1}}{(\Delta x)^2}.$$

With $s = k\Delta t/(\Delta x)^2$,

$$-sU_{j-1}^{m+1} + (1+2s)U_j^{m+1} - sU_{j+1}^{m+1} = U_j^m.$$

This is a tridiagonal linear system for the new time level. The implicit solve costs more per step but avoids the FTCS restriction on Δt for stability.

Crank–Nicolson scheme

Average the centered spatial difference between time levels m and $m+1$:

$$\frac{U_j^{m+1} - U_j^m}{\Delta t} = \frac{k}{2(\Delta x)^2}\left[\delta_{xx} U_j^{m+1} + \delta_{xx} U_j^m\right],$$

where $\delta_{xx} U_j = U_{j+1} - 2U_j + U_{j-1}$.

- Second-order accuracy in time and space for smooth solutions.
- Each time step requires a linear solve.
- Stable for all $s > 0$ for the standard linear heat equation, although very large time steps can still give poor accuracy.

Boundary conditions: Dirichlet and Neumann

Dirichlet condition: specify the solution value, e.g. $u(0,t) = 0$.

$$U_0^m = 0 \quad \text{at every time level.}$$

Neumann condition: specify a derivative, e.g. $u_x(0,t) = 0$ (insulated end).

$$u_x(0,t) \approx \frac{U_1^m - U_{-1}^m}{2\Delta x} = 0$$

can be implemented with a ghost point or a suitable one-sided formula. Boundary treatment must match the physical condition and can affect the stability analysis.

Heat equation vocabulary: the discrete problem

Partial difference equation (PDE in the textbook's terminology) An algebraic relation among approximate solution values at neighboring space-time mesh points. Here it is distinct from the original partial differential equation.

Local truncation error The residual obtained by substituting the exact smooth solution into the difference equation; it measures the formula's local approximation error.

Explicit scheme The next-time value is calculated directly from values already known at the current time level. FTCS is explicit.

Implicit scheme Unknown values at the new time level occur together, so each time step requires solving a system of equations. Crank–Nicolson is implicit.

Dimensionless mesh parameter $s = k\Delta t/(\Delta x)^2$ for the 1-D heat equation. Its value controls the FTCS update and stability.

Nonhomogeneous problem A problem with a source term and/or nonzero boundary data, e.g. $u_t = ku_{xx} + Q(x, t)$.

Heat equation vocabulary: stability and convergence

Stability Small perturbations in data or roundoff errors do not grow without bound under repeated time stepping. For a linear mode with amplification factor G, require $|G| \leq 1$.

Amplification factor The factor by which a discrete spatial mode is multiplied after one time step.

Fourier–von Neumann stability analysis Test the growth of Fourier modes by substituting a periodic mode into the linear difference equation.

Convergence As the mesh is refined, the numerical solution approaches the solution of the differential equation (under the appropriate assumptions).

Lax equivalence theorem For a well-posed linear initial-value problem and a consistent finite-difference scheme, stability is equivalent to convergence.

Eigenfunction / eigenvalue A spatial pattern preserved by a linear operator up to multiplication by a scalar; the scalar determines the mode's growth or decay in the discrete evolution.

Heat equation vocabulary: schemes and boundary treatment

Richardson scheme A centered-in-time, centered-in-space heat scheme. Although its formal truncation error is second order in both variables, the textbook shows that this scheme is unstable and should not be used.

Crank–Nicolson scheme Average the spatial second difference at times t_m and t_{m+1}. It is second-order accurate in time and space for a smooth solution and is stable for the standard linear heat problem, but requires a linear solve per step.

Neumann boundary condition Prescribes a normal derivative, such as $u_x(0,t) = g(t)$; physically, it can represent a specified heat flux.

Fictitious (ghost) point An auxiliary grid value just outside the physical domain, introduced to approximate a derivative boundary condition; eliminate it using the boundary relation.

Matrix form The FTCS update can be written $\mathbf{U}^{m+1} = A\mathbf{U}^m$ (for homogeneous fixed boundary data), making eigenvalues of A central to stability.

Heat equation: additional spectral and matrix terms

Discrete Fourier mode A grid-scale sine or complex exponential that is preserved in shape by the linear heat update and multiplied by an amplification factor each step.

Discrete orthogonality A finite-sum analogue of orthogonality of continuous sine functions; it allows initial grid data to be expanded in independent discrete modes.

Eigenvector expansion Write the grid solution as a linear combination of eigenvectors of the update matrix. Each coefficient evolves by repeated multiplication by its eigenvalue.

Gershgorin circle theorem Every eigenvalue of a matrix lies in at least one disk centered at a diagonal entry, with radius equal to the sum of the absolute values of the other entries in that row. It can bound eigenvalues when exact calculation is difficult.

Random-walk interpretation For suitable s, the FTCS heat update is a weighted average of a node and its neighbors, resembling a discrete diffusion or random-walk step. This is an interpretation, not a replacement for stability analysis.

Section 6.3: heat-equation checklist

Before running a heat computation:

1. Set $U_j^0 = f(x_j)$.
2. Impose the boundary values at every time step.
3. Compute $s = k\Delta t/(\Delta x)^2$.
4. For FTCS, ensure $s \leq 1/2$.
5. Update all interior nodes and check the result for nonphysical oscillations.
6. Refine the mesh and compare results when possible.

6.4 From one dimension to two dimensions

The two-dimensional heat equation is

$$u_t = k(u_{xx} + u_{yy}) = k\nabla^2 u.$$

Now the spatial mesh has nodes (x_i, y_j), and for a rectangular grid we often take $\Delta x = \Delta y = h$. At each interior point, heat diffusion depends on four neighboring points: left, right, up, and down.

The 2-D explicit heat update

Use forward time and the five-point Laplacian:

$$\frac{U_{i,j}^{m+1} - U_{i,j}^{m}}{\Delta t} = \frac{k}{h^2}\left(U_{i+1,j}^{m} + U_{i-1,j}^{m} + U_{i,j+1}^{m} + U_{i,j-1}^{m} - 4U_{i,j}^{m}\right).$$

Set $s = k\Delta t/h^2$:

$$\boxed{U_{i,j}^{m+1} = s(\text{four neighbors}) + (1-4s)U_{i,j}^{m}.}$$

Here "four neighbors" means the sum of the four neighboring values at time m.

2-D stability condition

A Fourier mode $e^{i(i\theta+j\phi)}$ has amplification factor

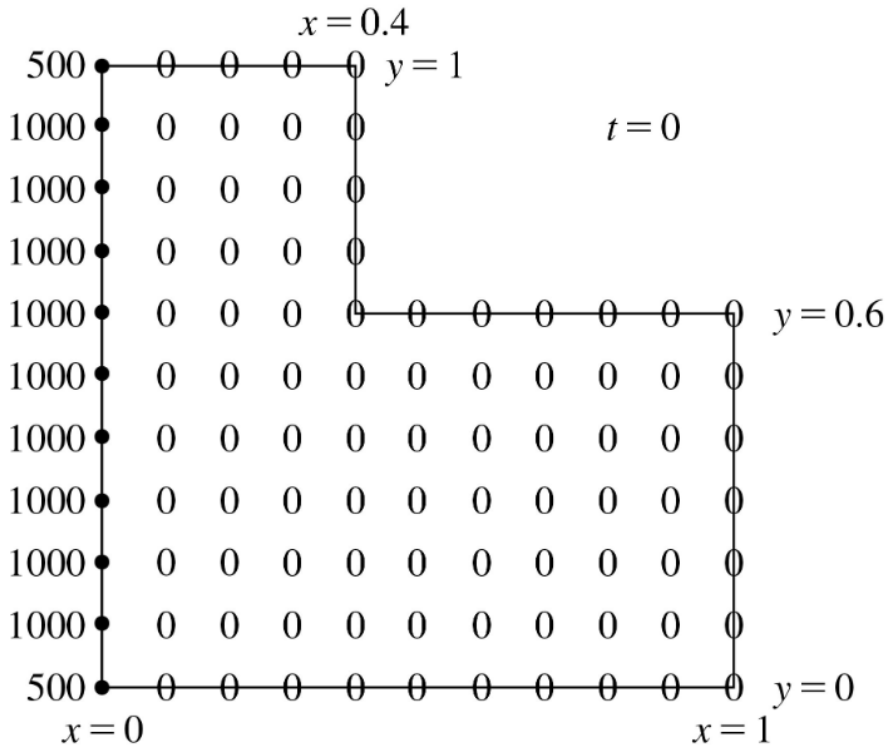
$$G = 1 + 2s(\cos\theta + \cos\phi - 2).$$

Since $-1 \le \cos\theta, \cos\phi \le 1$, the minimum is $1 - 8s$. Requiring $G \ge -1$ gives

$$\boxed{s = \frac{k\Delta t}{h^2} \le \frac{1}{4}.}$$

Thus the standard explicit 2-D heat method requires $\Delta t \le h^2/(4k)$ on a uniform square grid. The extra spatial direction tightens the restriction compared with 1-D.

Textbook example: initial data on an L-shaped region



The textbook example starts from zero temperature in the interior, fixes one boundary segment at a high temperature, and holds the remaining boundary at zero. The computation shows how heat diffuses into an irregular region. Cropped

from Haberman, Chapter 6, Fig. 6.4.1 (printed p. 254).

In-class example: one 2-D heat update

Question. Let $s = 1/8$ at an interior grid point. Its current value is $U_{i,j}^{m} = 10$, and its north, south, east, and west neighbors are $8, 6, 12, 14$. Use the 2-D explicit update to compute $U_{i,j}^{m+1}$. Is the selected s within the standard stability limit?

Irregular boundaries and practical issues

- The stencil is straightforward when the boundary aligns with grid points.
- An irregular boundary may cut through grid cells, complicating the boundary treatment.
- The textbook's L-shaped example uses contours to display how temperature enters the region.
- The 2-D method generalizes naturally to 3-D, but each added dimension increases computational work and tightens explicit stability restrictions.

Section 6.4: extending the heat method to two dimensions

For $u_t = k(u_{xx} + u_{yy})$ on a uniform square grid with $\Delta x = \Delta y = h$, the explicit update is

$$U_{i,j}^{m+1} = U_{i,j}^m + s\left(U_{i+1,j}^m + U_{i-1,j}^m + U_{i,j+1}^m + U_{i,j-1}^m - 4U_{i,j}^m\right), \quad s = \frac{k\Delta t}{h^2}.$$

- The center and four nearest neighbors form the 2-D five-point stencil.
- The standard explicit stability restriction on a square grid is $s \leq 1/4$ (compare $s \leq 1/2$ in 1-D).
- For unequal spacings, the x and y contributions have different weights; do not use the square-grid bound without adjustment.
- On an irregular region, boundary points and the grid geometry need careful treatment. The textbook's L-shaped example illustrates this issue.
- In 3-D, six nearest neighbors enter the standard Laplacian stencil, and explicit stability becomes more restrictive.

6.5 The one-dimensional wave equation

The wave equation is

$$u_{tt} = c^2 u_{xx}, \qquad 0 < x < L,$$

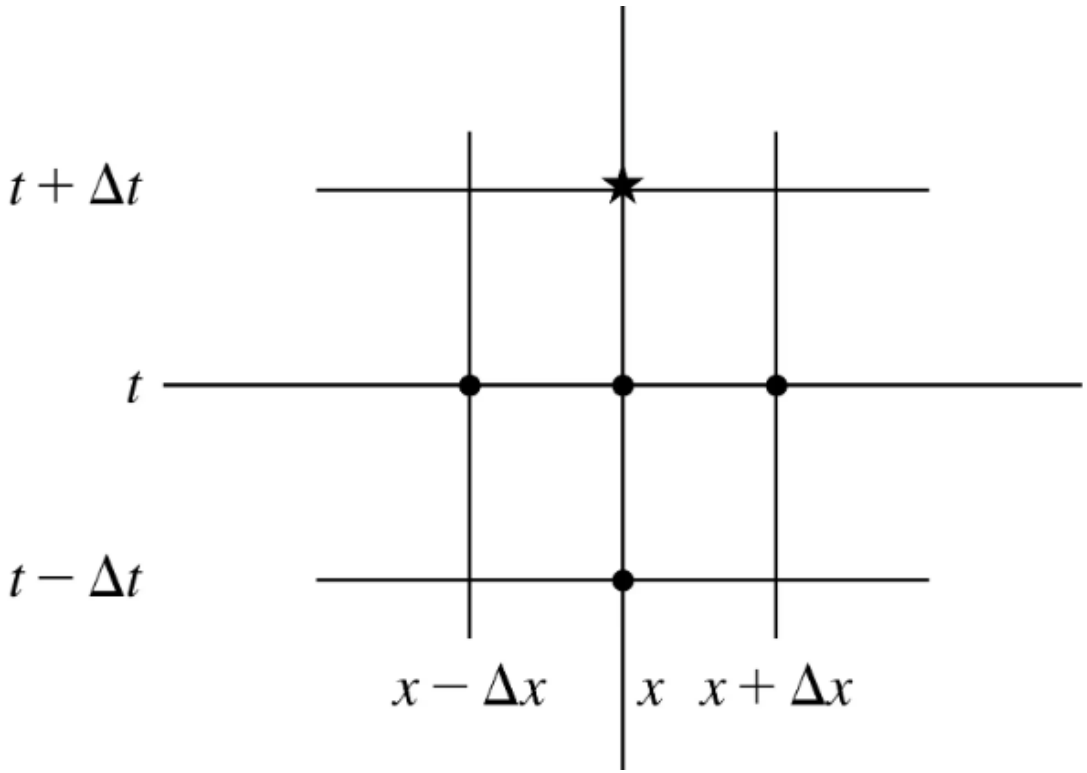
with initial displacement and velocity

$$u(x,0) = f(x), \qquad u_t(x,0) = g(x),$$

and boundary conditions such as $u(0,t) = u(L,t) = 0$.

- c is the wave speed.
- Two time derivatives mean that both displacement and velocity are needed initially.
- Unlike diffusion, waves can propagate without smoothing away all oscillations.

Textbook figure: wave-equation stencil



The centered wave scheme uses three spatial points at time t_m and the center spatial point at times t_{m-1} and t_{m+1}. Cropped from Haberman, Chapter 6, Fig. 6.5.1 (printed p. 257).

Derive the centered wave scheme

Approximate both second derivatives by centered differences:

$$\frac{U_j^{m+1} - 2U_j^m + U_j^{m-1}}{(\Delta t)^2} = c^2 \frac{U_{j+1}^m - 2U_j^m + U_{j-1}^m}{(\Delta x)^2}.$$

Let $r = c\Delta t/\Delta x$. Solving for the new time level gives

$$\boxed{U_j^{m+1} = 2U_j^m - U_j^{m-1} + r^2(U_{j+1}^m - 2U_j^m + U_{j-1}^m).}$$

The local truncation error is $\mathcal{O}((\Delta t)^2) + \mathcal{O}((\Delta x)^2)$ for a sufficiently smooth solution.

How to start the wave computation

The recurrence needs two time levels, U_j^0 and U_j^1. The displacement gives $U_j^0 = f(x_j)$, but the initial velocity must be used to calculate the first new row. Taylor-expand in time:

$$u(x_j, \Delta t) = u(x_j, 0) + \Delta t u_t(x_j, 0) + \frac{(\Delta t)^2}{2} u_{tt}(x_j, 0) + \mathcal{O}((\Delta t)^3).$$

Since $u_{tt} = c^2 u_{xx}$, use the PDE and the centered spatial difference to obtain the startup formula.

Wave startup formula

Using $u_t(x,0) = g(x)$ and centered space differences,

$$\boxed{U_j^1 = U_j^0 + \Delta t g(x_j) + \frac{r^2}{2}(U_{j+1}^0 - 2U_j^0 + U_{j-1}^0).}$$

This gives a second-order start under the smoothness assumptions used in the Taylor expansion. After the first step, use the three-level recurrence and impose boundary values.

Wave stability and the Courant number

Substitute a Fourier mode into the recurrence. The amplification factor satisfies a quadratic relation; requiring non-growing wave modes yields the Courant condition

$$\boxed{r = \frac{c\Delta t}{\Delta x} \leq 1, \qquad \Delta t \leq \frac{\Delta x}{c}.}$$

- For the heat equation, stable Δt scales like $(\Delta x)^2$.
- For the wave equation, stable Δt scales like Δx.
- The condition relates the numerical propagation speed to the physical wave speed.

In-class example: wave-equation step size

Question. For wave speed $c = 2$ and spatial spacing $\Delta x = 0.1$, find the largest time step permitted by the Courant condition. Then write the centered recurrence for an interior node and identify the time levels used.

Wave equation vocabulary and interpretation

Wave speed The physical propagation speed c in $u_{tt} = c^2 u_{xx}$.

Three-level scheme The centered wave update uses values at t_{m-1}, t_m, and t_{m+1}; two time levels are involved in the recurrence.

Startup step The first computed level cannot be obtained from the recurrence alone. Use both $u(x, 0) = f(x)$ and $u_t(x, 0) = g(x)$ to derive U_j^1.

Courant number $r = c\Delta t/\Delta x$, the ratio of physical wave travel over one time step to one spatial mesh width.

Courant stability condition For the standard centered scheme on a uniform 1-D grid, $r \leq 1$, equivalently $\Delta t \leq \Delta x/c$.

Discrete dispersion The numerical relation between spatial wave number and temporal frequency generally differs from the continuous wave equation; resolved low-frequency modes approximate the continuous frequency better.

Wave-equation checklist

1. Initialize displacement: $U_j^0 = f(x_j)$.
2. Use initial velocity to compute U_j^1.
3. Check $r = c\Delta t/\Delta x \leq 1$ for the standard explicit scheme.
4. Apply boundary values at every time level.
5. Advance using the centered three-level recurrence.

A stable result may still be inaccurate if the mesh is too coarse to resolve the wave.

6.6 Laplace's equation: a steady-state problem

Laplace's equation in a region R is

$$\nabla^2 u = u_{xx} + u_{yy} = 0 \quad \text{in } R,$$

with prescribed boundary values. It models steady temperature in a source-free region, among other applications.

- There is no time variable in the equation.
- Boundary data determine the interior field.
- The finite-difference approximation produces a system of algebraic equations.

From Laplace's equation to the five-point equation

Use the centered approximation for each second derivative:

$$\frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{h^2} + \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{h^2} = 0.$$

Multiply by h^2 and rearrange:

$$\boxed{U_{i,j} = \frac{1}{4}\left(U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1}\right).}$$

Each interior value equals the average of its four nearest neighbors.

The discrete mean-value property

The five-point equation says the value at an interior node is the average of the four adjacent values.

- This resembles the mean-value property of harmonic functions.
- Interior values are constrained by surrounding values and the boundary data.
- The boundary values are known; the interior nodal values are unknown.
- Assembling all interior equations gives a sparse linear system $A\mathbf{U} = \mathbf{b}$.

Solving the Laplace system

In principle, one can solve $A\mathbf{U} = \mathbf{b}$ by Gaussian elimination. For a large grid, direct solution may be expensive, so iterative methods are often used.

1. Start with a guess for every interior value.
2. Update the guess using the five-point equation.
3. Repeat until the update or residual is sufficiently small.

The textbook discusses Jacobi, Gauss–Seidel, and successive over-relaxation (SOR).

Jacobi iteration

Every new value uses only values from the previous iteration:

$$U_{i,j}^{(n+1)} = \frac{1}{4}\left(U_{i+1,j}^{(n)} + U_{i-1,j}^{(n)} + U_{i,j+1}^{(n)} + U_{i,j-1}^{(n)}\right).$$

- All new values can be computed independently from the old grid.
- It is simple to implement and parallelize.
- It often converges slowly, especially on fine grids.

Boundary values stay fixed throughout the iteration.

Gauss–Seidel iteration

Gauss–Seidel updates nodes sequentially, using new values as soon as they are available. For one common left-to-right, bottom-to-top sweep,

$$U_{i,j}^{(n+1)} = \frac{1}{4}\left(U_{i-1,j}^{(n+1)} + U_{i+1,j}^{(n)} + U_{i,j-1}^{(n+1)} + U_{i,j+1}^{(n)}\right).$$

The exact old/new terms depend on the sweep order. Compared with Jacobi, Gauss–Seidel usually converges faster but is less directly parallel.

Successive over-relaxation (SOR)

Let $U_{i,j}^{GS}$ denote the Gauss–Seidel update. SOR takes a relaxed step:

$$U_{i,j}^{new} = (1-\omega)\,U_{i,j}^{old} + \omega\,U_{i,j}^{GS}.$$

- $\omega = 1$ gives Gauss–Seidel.
- $0 < \omega < 1$ under-relaxes; $1 < \omega < 2$ over-relaxes.
- For the standard model problem, a suitable ω can substantially accelerate convergence; the best value depends on geometry and mesh.

How do we decide when to stop iterating?

Two common diagnostics are:

$$\|\mathbf{U}^{(n+1)} - \mathbf{U}^{(n)}\|_\infty < \varepsilon$$

or the residual test

$$\|A\mathbf{U}^{(n)} - \mathbf{b}\|_\infty < \varepsilon.$$

A small update does not always imply a small error, so residuals and mesh-refinement checks are valuable. The tolerance ε should be chosen in relation to the problem's scale and desired accuracy.

In-class example: one Laplace iteration

Question. An interior node has north, south, east, and west values $8, 4, 6, 2$. Compute its Jacobi update. Then explain how the Gauss–Seidel update would change if the newest west and south values were already available in the current sweep.

Iteration methods: qualitative comparison

Method	Values used	Main feature
Jacobi	Previous iteration only	Simple and parallel; often slow.
Gauss–Seidel	New values as available	Usually faster per iteration.
SOR	Relaxed Gauss–Seidel update	Can reduce iteration count if ω is well chosen.

The textbook also notes that other improved methods, including alternating-direction implicit methods, have been developed.

Laplace equation vocabulary: steady state and iteration

Elliptic equation A PDE class exemplified by $\nabla^2 u = 0$; here there is no time variable, and boundary data determine the interior steady state.

Discrete mean-value property The five-point Laplace equation says each interior grid value equals the average of its four nearest neighbors.

Discrete maximum/minimum principle For the standard Laplace discretization, interior values cannot exceed the largest boundary value or fall below the smallest boundary value.

Iterative method Start from a guess and repeatedly update it to approach the solution of the discrete linear system.

Jacobi iteration Every new value is computed from the neighboring values of the previous iterate; all updates use the same old grid.

Gauss–Seidel iteration Use newly updated values immediately during a sweep; for the standard Laplace problem it typically converges faster than Jacobi.

Successive over-relaxation (SOR) Modify a Gauss–Seidel update by a relaxation parameter ω to accelerate convergence; for the standard model problem, the usual convergent range is $0 < \omega < 2$.

Laplace equation vocabulary: residual and stopping

Residual The mismatch when a current iterate is substituted into the discrete equations, e.g. $R_{i,j} = U_{i+1,j} + U_{i-1,j} + U_{i,j+1} + U_{i,j-1} - 4U_{i,j}$.

Stopping criterion A practical rule for ending iterations, such as requiring the maximum update or residual norm to be below a chosen tolerance.

Convergence rate How quickly the iteration error decreases. For fine grids, Jacobi can be very slow; Gauss–Seidel often improves the rate, while a suitable SOR parameter can improve it further.

Boundary values In the discrete system, known boundary values contribute to the right-hand side; only interior grid values are unknowns.